



Course: Linear Algebra and Optimization

Course Code: 23MA4BSLAO

Unit 4: Eigenvalues and Eigenvectors

I. Eigenvalues and Eigenvectors of a linear transformation:

A scalar λ is called an eigenvalue of a linear transformation $T: V \rightarrow V$ if there is a non-zero vector X such that $T(X) = \lambda X$. The vector X is called an eigenvector of the linear transformation T corresponding to λ .

The set of all eigenvectors of λ along with the zero-vector form subspace called eigenspace of λ .

Any linear transformation $T: V \rightarrow V$ then λ is an eigenvalue of the matrix of the linear transformation, A and X is the corresponding eigenvector.

Theorem: The eigenvalues and eigenvectors of a matrix $A \in M_{n \times n}$ satisfies the following:

- i. An eigenvalue of A is a scalar λ such that $|A - \lambda I| = 0$.
- ii. The eigenvectors of A corresponding to λ are the nonzero solutions of $(A - \lambda I)x = 0$ i.e. the null space of $A - \lambda I$.
- iii. Characteristic polynomial of A is $|A - \lambda I| = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0$.

Theorem: Let A be a square matrix. Then the following are equivalent.

- a) A scalar λ is an eigen value of A .
- b) The matrix $M = A - \lambda I$ is singular.
- c) The scalar λ is a root of the characteristic polynomial.

Note:

- i. The Characteristic Equation of 2 by 2 matrix: $\lambda^2 - \text{trace}(A)\lambda + |A| = 0$.
- ii. For 3 by 3 matrix: $-\lambda^3 + \text{trace}(A)\lambda^2 - (\sum_{i=j} M_{ij})\lambda + |A| = 0$ where M_{ij} is the principal minors of A .

II. Properties of eigenvalues and eigen vectors

- 1) There are infinitely many eigenvectors corresponding to each eigenvalue but for a given eigenvector the eigenvalue is unique.
- 2) Eigenvectors of a matrix A corresponding to distinct eigenvalues are linearly independent.
- 3) Trace of A is equal to the sum of the eigenvalues of A .
- 4) Determinant of A is equal to product of eigenvalues A .
- 5) A and A^T has same eigenvalues.
- 6) Eigenvectors of distinct eigenvalues of real symmetric matrices are orthogonal.
- 7) If A is a square matrix then the following statements are equivalent:
 - a) The eigenvalues are non-zero ($|A| \neq 0$).
 - b) A^{-1} exists and $1/\lambda$ is the eigenvalue of A^{-1}
- 8) Zero is one of the eigenvalue if and only if A is singular.
- 9) If λ is an eigenvalue of A then
 - a) λ^m is the eigenvalue of A^m .
 - b) $k\lambda$ is the eigenvalue of kA .



III. Cayley-Hamilton Theorem:

Statement: Every Square Matrix satisfies its own characteristic equation.

Apply Cayley-Hamilton Theorem to find or Compute using minimal conventional operations.

1) A^{-1} if $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$.

2) A^4 given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$.

3) A^{-1} given $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Also Compute A^8 .

4) Find A^{-1} , A^{-2} , A^{-3} if $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$.

5) Given $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ prove that $(4M - 3I)^{-2} = M^{-4}$.

6) If $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ then Evaluate the expression $A^8 - A^7 + A^2 + I$.

7) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ find M^{-2} .

8) Solve the following system of linear equations $AX=B$ by finding the inverse of the coefficient matrix using the Cayley-Hamilton Theorem:

$$\begin{aligned} 2x - y + z &= 2 \\ 3x + y - 2z &= 9 \\ -x + 2y + 5z &= -5 \end{aligned}$$

9) If $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 4 & 0 & 1 \\ 3 & 1 & 5 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and $\lambda_1 = 1$ and $\lambda_2 = 2$ then is the matrix invertible?

ANS: $\lambda_3 = 4, \lambda_4 = 5$.

10) If $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 1 & 2 & 0 & 5 \\ 0 & 4 & 2 & 0 \\ 2 & 4 & 0 & 0 \end{bmatrix}$ and $\lambda_1 = -4$ and $\lambda_2 = 2$ then is the matrix invertible?

ANS: $\lambda_3 = 0, \lambda_4 = 7$.

IV. Eigen Spaces:

Theorem: (The eigenspace of A corresponding to λ)

If $T: V \rightarrow V$ is a linear transformation then the subset of all eigenvectors corresponding to λ along with the zero vector is a subspace of V and is called the eigen space of λ denoted by $E(\lambda)$.

Note:

1. If an eigenvalue λ occurs as a multiple root (k times) of the characteristic polynomial, then λ has **algebraic multiplicity** (A.M.) k .



2. The number of independent eigenvectors corresponding to a given eigenvalue is the geometric multiplicity (G.M.) and hence the dimension of the eigenspace of an eigenvalue λ of A is the **geometric multiplicity** of λ .
3. $0 < G.M. \leq A.M.$ and if $A.M. = 1$ then $G.M. = 1$.
4. The eigenvalue λ is said to be defective if the geometric multiplicity of λ is less than its algebraic multiplicity.
5. An $n \times n$ matrix A is said to be defective if it doesn't have n linearly independent eigenvectors. Equivalently, an $n \times n$ matrix A is said to be defective if it has a defective eigenvalue.

Obtain algebraic and geometric multiplicity of the eigenvalues for the following linear transformations (or Matrices) by finding the eigenspaces and hence identify the defective eigenvalues/matrices.

- 1) $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$.
- 2) $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.
- 3) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (3x + 3y, x + 5y)$.
- 4) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y + z, x - y + z, x - 2y + 2z)$
- 5) $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$.
- 6) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.
- 7) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (2x - 5y, 2y, -x + 4y + z)$.
- 8) $T: P_1 \rightarrow P_1$ defined by $T(at + b) = (3a + 5b)t - (2a + 4b)$.
- 9) $T: P_2 \rightarrow P_2$ defined by $T(at^2 + bt + c) = (2a - c)t^2 + (2a + b - 2c)t + (-a + 2c)$.
- 10) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T = \begin{bmatrix} 2 & 0 & 0 \\ -1 & -1 & 9 \\ 0 & -1 & 5 \end{bmatrix}$.
- 11) $A = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 1 \\ -2 & 2 & 1 \end{bmatrix}$.
- 12) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and eigenvalues of M are $\lambda = 0, 2$. Find the algebraic and geometric multiplicity of each λ and hence find eigenspace corresponding to each eigenvalue of $A = M^2 + \frac{1}{2}M$.
- 13) Let $f(t) = t^3 + 2t^2 + 3$ be the polynomial of degree 3 and X is the eigen vector of A corresponding to eigen value λ . Prove that X is the eigenvector of $f(A)$ corresponding to eigenvalue $f(\lambda)$.
- 14) $D: V \rightarrow V$ defined by $D(f) = \frac{df}{dt}$, where V is the space of functions with basis $S = \{\sin t, \cos t\}$.
- 15) Let $T: S \rightarrow S$ is defined by $T(A) = BA + AB$, where $A \in S$, and $S = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Show that the distinct eigen values of T are $\lambda = 0, 2, -2$.

**Minimal polynomial and Jordan Canonical Forms:**

Monic Polynomial: A polynomial $f(t)$ is monic if its leading coefficient equals one.

Minimal polynomial: Let A be a square matrix. Let $J(A)$ denote the collection of polynomials for which A is a root that is, for which $f(A) = 0$. The set is not empty [because the Cayley-Hamilton theorem tells us that the characteristic polynomial belongs to $J(A)$]. Let $m(t)$ denote the monic polynomial of lowest degree in $J(A)$. We call $m(t)$ the minimal polynomial of the matrix.

- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2, \dots, A_n$. Then the characteristic polynomial of M is the product of the characteristic polynomials of the diagonal blocks A_i .
- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2, \dots, A_n$. Then the minimal polynomial of M is equal to the least common multiple of the minimal polynomials of the diagonal blocks A_i .

1) Find the minimal polynomial of

a) $\begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}$.

b) $\begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$.

c) $\begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}$.

d) $\begin{bmatrix} 3 & 1 & -1 \\ 2 & 4 & -2 \\ -1 & -1 & 3 \end{bmatrix}$.

2) Find the characteristic and minimal polynomial of following

i) $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix}$.

ii) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$.

iii) $\begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}$.

- 3) Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Show that A and B have different characteristic polynomials (and so are not similar) but have the same minimal polynomial. Thus, nonsimilar matrices may have the same minimal polynomial.



V. Jordan Canonical Forms:

Let $T: V \rightarrow V$ be a linear operator whose characteristic and minimal polynomials are, respectively, $f(t) = (t - \lambda_1)^{n_1} \dots (t - \lambda_r)^{n_r}$ and $m(t) = (t - \lambda_1)^{m_1} \dots (t - \lambda_r)^{m_r}$.

where the λ_i are distinct scalars. Then T has a block diagonal matrix representation J in which each diagonal entry is a Jordan block $J_{ij} = J(\lambda_i)$. For each λ_i , the corresponding J_{ij} have the following properties:

(i) There is at least one J_{ij} of order m_i ; all other J_{ij} are of order $\leq m_i$.

(ii) The sum of the orders of the J_{ij} is n_i .

(iii) The number of J_{ij} equals the geometric multiplicity of λ_i .

(iv) The number of J_{ij} of each possible order is uniquely determined by T .

- 1) Find the Jordan canonical forms of T given by characteristic and minimal polynomial of T respectively $f(t) = (t-2)^4(t-5)^3$ and $m(t) = (t-2)^2(t-5)^3$.
- 2) Write Jordan canonical form or blocks if $f(t) = (t-2)^5$ is characteristic polynomial and $m(t) = (t-2)^2$ is the minimal polynomial.
- 3) Write Jordan canonical form or blocks if $f(t) = (t-2)^2(t-3)$ is characteristic polynomial and $m(t) = (t-2)(t-3)$ is the minimal polynomial.
- 4) Find the Jordan canonical forms of T given by characteristic and minimal polynomial of T respectively $f(t) = (t+1)^5(t-4)^2$ and $m(t) = (t+1)^3(t-4)^2$.
- 5) Write Jordan canonical form or blocks if $D(t) = (t+3)^6$ is characteristic polynomial and $m(t) = (t+3)^2$ is the minimal polynomial.
- 6) Write Jordan canonical form or blocks if $f(t) = (t-4)^4$ is characteristic polynomial and $m(t) = (t-4)^2$ is the minimal polynomial.

7) Find the Jordan canonical form of $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 0 \\ 1 & -2 & 3 \end{bmatrix}$.

8) Find the Jordan canonical form of $\begin{bmatrix} 2 & 1 & 0 & 0 \\ -1 & 4 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 4 \end{bmatrix}$.